

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VII

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Hash Table:

Write a program to:

- a. Implement a hash table with separate chaining for collision handling.

CODE :

```
size = 5
table = [[]for _ in range(size)]

def insert(key,value):
    table[(key%size)].append((key,value))

insert(10,"SAHIL")
insert(15,"KRISH")
insert(20,"DHRUV")
insert(32,"AKKU")
insert(33,"BALI")

print(table)
```

CODE :

```
| [(10, 'SAHIL'), (15, 'KRISH'), (20, 'DHRUV')], [], [(32, 'AKKU')], [(33, 'BALI')], []]
```

- b. Store and retrieve data from the hash table.

CODE :

```
size = 5
table = [[]for _ in range(size)]

def insert(key,value):
    table[(key%size)].append((key,value))
def search(key):
    for k,v in table[key%size]:
        if k==key:
            return v
    return "Not Found"

insert(10,"SAHIL")
insert(15,"KRISH")
insert(20,"DHRUV")
insert(32,"AKKU")
insert(33,"BALI")

print(table)

key = int(input("Enter key : "))
print(search(key))

print("Abhishek S025")
```

CODE :

```
[[ (10, 'SAHIL'), (15, 'KRISH'), (20, 'DHRUV') ], [], [(32, 'AKKU')], [(33, 'BALI')], []]
Enter key : 32
AKKU
Abhishek S025
```